

Code:

```
#include <stdio.h> #include <stdlib.h>

#define MAXM 10005

long long count_mod[MAXM];

int main() { int N, M; if (scanf("%d %d", &N, &M) != 2) return 0;

for (int idx = 0; idx < N; idx++) {
    long long val;
    scanf("%lld", &val);
    count_mod[val % M]++;
}

long long total_triplets = 0;

// Case 1: All three remainders equal (r, r, r) -> 3*r = 0 (mod M)
for (int r = 0; r < M; r++) {
    if ((3LL * r) % M == 0) {
        long long c = count_mod[r];
        if (c >= 3) {
            total_triplets += c * (c - 1) * (c - 2) / 6;
        }
    }
}

// Case 2: Two remainders equal (r1, r1, r2) with r1 != r2 -> (2*r1 + r2) = 0 (mod M)
for (int r1 = 0; r1 < M; r1++) {
    long long c1 = count_mod[r1];
    if (c1 < 2) continue;

    int r2 = (M - (2 * r1) % M) % M;
    if (r1 != r2) {
        total_triplets += (c1 * (c1 - 1) / 2) * count_mod[r2];
    }
}

// Case 3: All three remainders distinct (i < j < k) with (i + j + k) = 0 (mod M)
for (int i = 0; i < M; i++) {
```

```

        if (count_mod[i] == 0) continue;
        for (int j = i + 1; j < M; j++) {
            if (count_mod[j] == 0) continue;
            int k = (M - (i + j) % M) % M;
            while (i < k) {
                if (j < k) {
                    total_triplets += count_mod[i] * count_mod[j] *
count_mod[k];
                }
                break;
            }
        }
    }

printf("%lld\n", total_triplets);
return 0;

}

```

Input:

```

10 5
1 10 4 3 2 5 0 19 5 1

```

Output:

